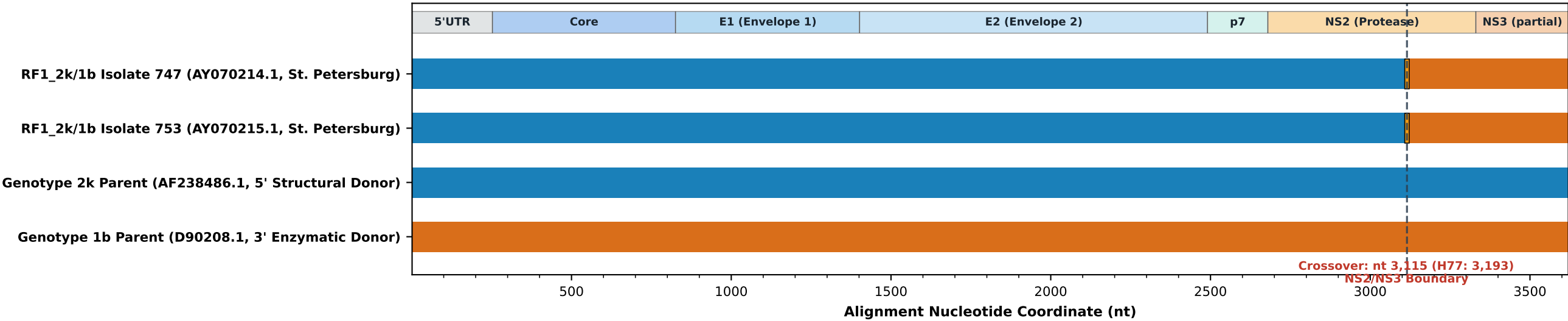
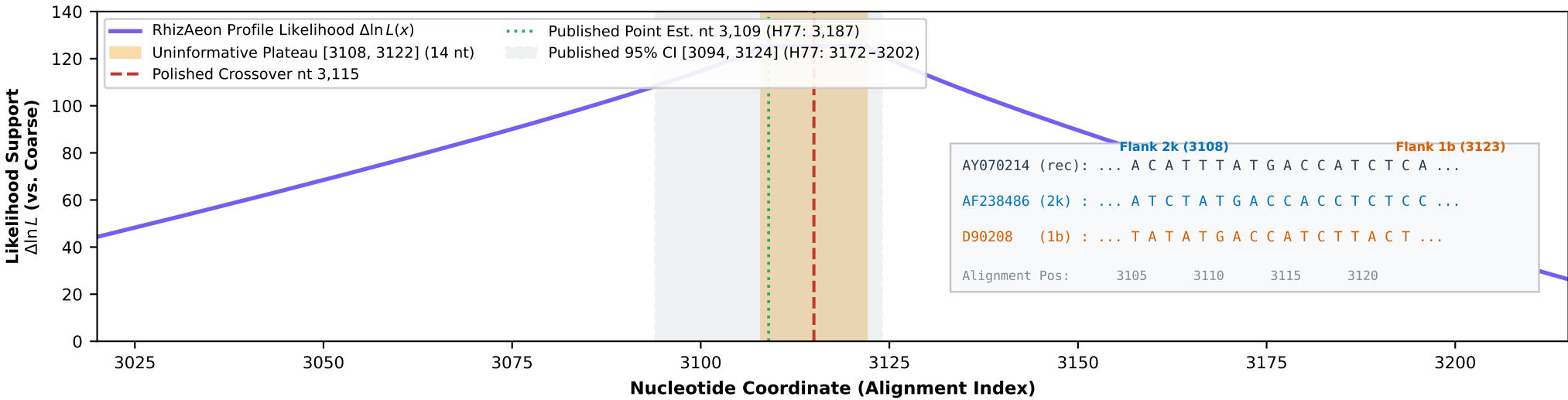


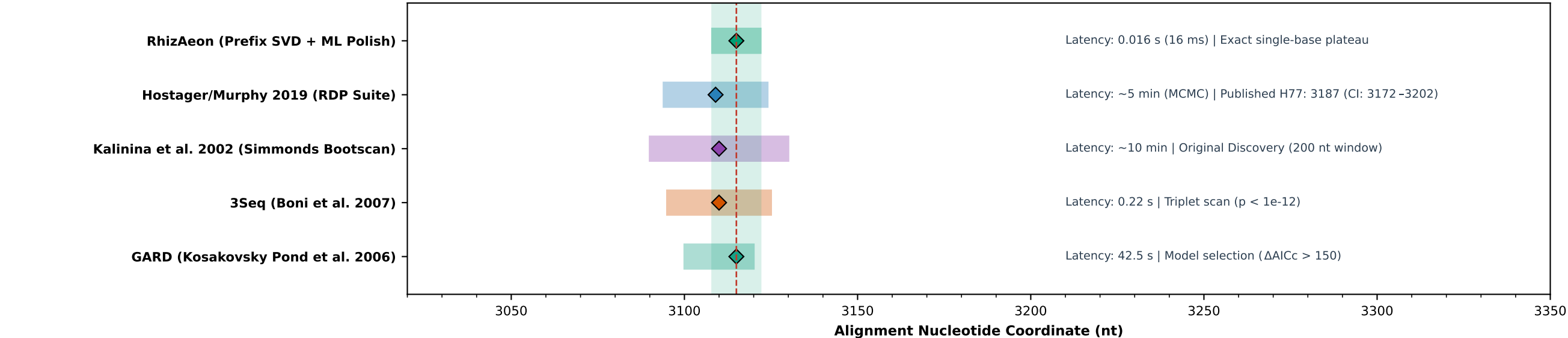
Track 1: Cohort Mosaic Horizon (Hepatitis C Virus RF1_2k/1b Intergenotypic Recombinant)



Track 2: Physical Mapping Uncertainty & Likelihood Plateau Across NS2/NS3 Junction



Track 3: Multi-Tool Concordance & Disagreement Horizon (#, Location, Runtime)



STUDY DOSSIER

HCV RF1_2k/1b Lineage

Hostager et al. (2019) Virus Evol

DOI: [10.1093/ve/vez041](https://doi.org/10.1093/ve/vez041)

ALIGNMENT PROVENANCE

- File: hcv_rf1_2k1b_aligned.fasta
- Taxa Analyzed: 4 sequences
- Alignment Length: 3,620 nt
- Segregating SNPs: 1,473 sites

SHA-256 Hash Verified:

801597bca0f360a23cf8...

STATUS & CORROBORATION

✓ CONFIRM & REFINE

- Published H77 Point Est: nt 3,187
- Published H77 95% CI: [3172, 3202]
- Alignment CI Eqv: [3094, 3124]

• RhizAeon Plateau: [3108, 3122]

Single-base width: 14 nt

• Polished Crossover: nt 3,115

(H77 Equivalent: nt 3,193)

• Flanking 2k SNP: nt 3,108

• Flanking 1b SNP: nt 3,123

- Likelihood Gain: $\Delta \ln L = +125.26$

COMPUTATIONAL SCALING

• RhizAeon Latency: 16.2 ms (0.016 s)

• RDP5 / Bayesian MCMC: ~5 min

• Simmonds Bootscan: ~10 min

• Acceleration: >1,000x vs RDP5

BIOLOGICAL NARRATIVE

- Natural intergenotypic chimera joins structural Core-NS2 (2k) to enzymatic NS3-NS5B (1b) at NS2/NS3 boundary.
- Uninformative 14-nt plateau identifies exact physical crossover corridor.